

Contact Information

Name Saptarshi Pal
Email spal@math.harvard.edu
Others links Google Scholar, ResearchGate

Research Interests

(a) Evolutionary of Cooperation, (b) Evolutionary Game Theory, (c) Mathematical Biology.

Academics

March 2024– **Post-Doctoral Fellow**,
Department of Mathematics, Harvard University.
Spring 2026 **Lecturer**, *Math 243: Evolutionary Dynamics*,
Harvard University.
2020–2024 **PhD in Mathematics and Natural Sciences**, *magna cum laude*,
Max Planck Institute for Evolutionary Biology.
Supervisor: Dr. Christian Hilbe
2018–2020 **Masters in Mathematics**,
University of Waterloo.
Supervisors: Dr. Chris T. Bauch and Dr. Madhur Anand
2014–2018 **Bachelor in Engineering**,
Jadavpur University, Kolkata, India.

Publications

- 2025 S. Pal, A. Mallela, C. Hilbe, L. Pracher, C. Wei, F. Fu, S. Schnell, M A Nowak.
"Strategies of cooperation and defection in five large language models" *pre-print*, (link).
- 2025 Philip LaPorte, Lenz Pracher, Saptarshi Pal
"From simultaneous to leader-follower play in direct reciprocity" *PNAS Nexus*, (link).
- 2025 Saptarshi Pal, Mayeul Lambert, Martin A. Nowak.
"Stabilizing unconditional cooperation" *Proceedings of Royal Society A*, (link).
- 2024 Saptarshi Pal, Nikoleta E. Glynatsi, Christian Hilbe.
"The co-evolution of direct, indirect and generalized reciprocity" *in review at Nature Communications pre-print at arxiv* (link).
- 2023 Marta C. Couto and Saptarshi Pal.
"Introspection dynamics in asymmetric multiplayer games" *Dynamic Games and Applications*, (link).
- 2022 Saptarshi Pal and Christian Hilbe.
"Reputation effects drive the joint evolution of cooperation and social rewarding".
Nature Communications, (link).
- 2021 Saptarshi Pal, Chris T. Bauch, and Madhur Anand.
"Coupled social and land use dynamics affect dietary choice and agricultural land-use extent"
Communications Earth & Environment, (link).

Teaching and Academic Supervisions

- Spring 2026 Lecturer at Harvard Math, Math 243: Evolutionary Dynamics
- Fall 2024 Supervised masters student Lenz Pracher at Harvard Math.
- Spring 2024 Supervised intern Mayeul Lambert at Harvard Math.
- Spring 2024 Lectures at Math 243, Harvard University.
- 2022 Lectures on Evolutionary Game Theory at the University of Kiel.
- 2022 Teaching Assistant for Math-WS21, Universität zu Lübeck.
Mathematical Biology in action during the Corona pandemic
- 2018-2020 Teaching Assistant at Faculty of Mathematics, University of Waterloo

Professional Activities

- 2021-Present **Peer reviewing** at Science, PNAS, Nature Communications, PNAS Nexus, Nature Sustainability, Dynamic Games and Application, PLOS One.
- 2023 **Organizer**, Lecture series on adaptive dynamics by Dr. Eva Kisdi at MPI Plön.

Others

- 2026(Conf.) Upcoming talk title: Strategies of cooperation and defection in five large language models, NITMB, Chicago, May, 2026.
- 2026(Conf.) Talk title: From simultaneous to leader-follower play in direct reciprocity, AMS Sectional Meeting, Boston, March 2026.
- 2023(Conf.) Talk title: A model of plausible deniability, Learning, Evolution and Games, Amsterdam.
- 2022(School) The Helsinki Summer School on Mathematical Ecology and Evolution.
- 2022(Conf.) Title: Reputation facilitates the co-evolution of cooperation and social rewarding
EHBEA, Max Planck Institute for Evolutionary Anthropology, Leipzig (Poster)
ICSD, International Conference on Social Dilemmas (Talk)
- 2022 (School) Toulouse Summer School in Quantitative Social Sciences, IAST and Toulouse School of Economics
- 2017 (Intern) MPI for Dynamics and Self Organization, Göttingen. Supervisor: Dr. Marc Timme.

Programming

PYTHON, MATHEMATICA, MATLAB, $\text{\LaTeX}$